

MCMC for Chemical Transformation Kinetics

1. Introduction

Environmental fate modelling requires robust parameter estimation in the face of sparse, noisy measurements. Classical least-squares regression yields single-point estimates but cannot quantify the full uncertainty that propagates from data noise into model predictions. Markov Chain Monte Carlo (MCMC) simulation overcomes this limitation by constructing the complete posterior distribution of model parameters, enabling rigorous probabilistic inference.

This report documents the design, implementation, and results of an MCMC-based analysis for a two-compartment chemical degradation model (Parent $\rightarrow$ Metabolite $\rightarrow$ Sink). Two regulatory scenarios are examined: Scenario A, where both the parent degradation rate k_p and the metabolite degradation rate k_m are non-zero, and Scenario B, where $k_m = 0$ (the metabolite is persistent). A central objective is to demonstrate that standard normal (Laplace) approximations can fail near physical boundaries — specifically the non-negativity constraint on rate constants.

2. Theoretical Background

2.1 Kinetic Model

The parent-metabolite system is governed by first-order coupled ODEs:

$$\begin{aligned}dP/dt &= -k_p \cdot P(t) \\dM/dt &= c \cdot k_p \cdot P(t) - k_m \cdot M(t)\end{aligned}$$

The analytical solution for the parent is $P(t) = M_0 \cdot \exp(-k_p \cdot t)$. For the metabolite when $k_p \neq k_m$: $M(t) = c \cdot M_0 \cdot k_p \cdot [\exp(-k_m \cdot t) - \exp(-k_p \cdot t)] / (k_p - k_m)$. A limiting (L'Hôpital) form applies when $k_p \approx k_m$. The four model parameters are: k_p (parent rate, d^{-1}), k_m (metabolite rate, d^{-1}), M_0 (initial mass, g/L), and c (molar conversion fraction, $0 \leq c \leq 1$).

2.2 Bayesian Framework & Log-Posterior

Under Gaussian measurement noise with unknown standard deviations σ_p (parent) and σ_m (metabolite), the log-posterior is:

$$\log \pi(\theta | y) = -\frac{1}{2} \sum [(y_{pi} - P_i)^2 / \sigma_p^2] - n \cdot \log(\sigma_p) - \frac{1}{2} \sum [(y_{mi} - M_i)^2 / \sigma_m^2] - n \cdot \log(\sigma_m)$$

Noise parameters are sampled on the log scale ($\log \sigma$) to enforce positivity. Flat (improper) priors are used within physical bounds: $k_p > 0$, $k_m \geq 0$, $M_0 > 0$, $0 \leq c \leq 1$.

2.3 Metropolis–Hastings Algorithm

The Metropolis–Hastings (MH) algorithm generates samples from the posterior by proposing moves from a symmetric Gaussian kernel and accepting or rejecting via the ratio of posterior densities:

$$\alpha = \min(1, \pi(\theta^* | y) / \pi(\theta_t | y))$$

For a 6-dimensional parameter vector, target acceptance rates lie in the range 15–40%. Proposal standard deviations were tuned empirically. A burn-in period discards transient samples prior to posterior inference.

3. Methods & Implementation

3.1 Data Generation

Synthetic observations were generated at 13 time points ($t = 0, 5, 10, 20, 30, 50, 70, 100, 150, 200, 250, 300, 365$ days) by adding 5% relative Gaussian noise to the analytical model solution. True parameter values are listed in Table 1.

Table 1. True parameter values used to generate synthetic data.

Parameter	Symbol	Scenario A	Scenario B
Parent rate constant	k_p (d ⁻¹)	0.058	0.058
Metabolite rate constant	k_m (d ⁻¹)	0.023	0.000
Initial mass	M_0 (g/L)	194	194
Molar conversion fraction	c (–)	0.224	0.224
Relative noise level	σ_{rel}	5%	5%

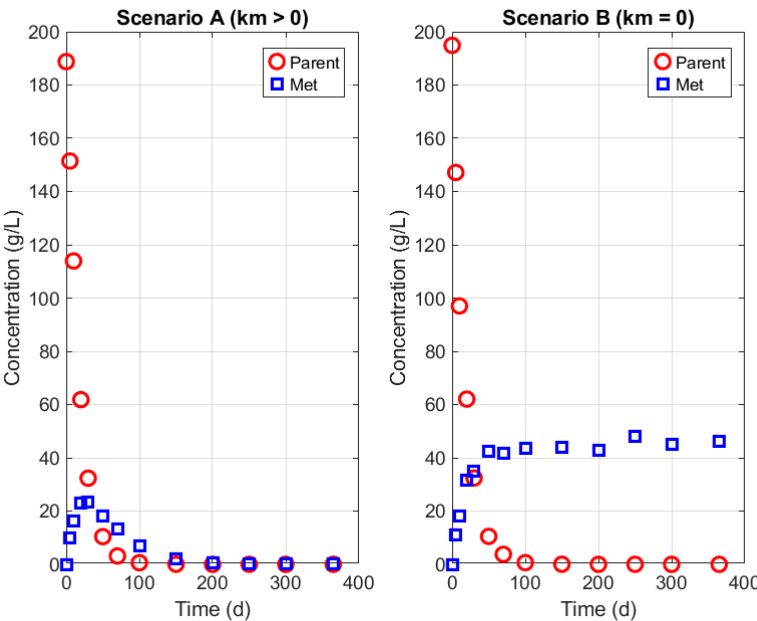


Figure 1. Synthetic concentration–time data for Parent (red circles) and Metabolite (blue squares) under Scenario A (left) and Scenario B (right).

3.2 MCMC Configuration

Key sampler settings are summarised in Table 2. Proposal standard deviations were chosen by pilot runs to achieve acceptance rates in the 15–45% range.

Table 2. MCMC sampler configuration.

Setting	Value
Total iterations	200,000
Burn-in period	50,000
Thinning factor	10
Effective posterior samples	15,000 per scenario
Proposal distribution	Gaussian (isotropic)
Proposal SD k_p	0.0005 d ⁻¹
Proposal SD k_m	0.0002 d ⁻¹
Proposal SD M_0	0.5 g/L
Proposal SD c	0.002
Proposal SD $\log\sigma_p, \log\sigma_m$	0.015

4. Results

4.1 Scenario A — Non-zero Metabolite Rate ($k_m = 0.023 \text{ d}^{-1}$)

The MCMC chain for Scenario A achieved an acceptance rate of 42.9%, within the recommended range for a 6-dimensional random walk. The posterior distributions for all four kinetic parameters are well-resolved and approximately Gaussian, closely centred on the true values. The 5th-percentile of the k_m posterior is 0.0228 d⁻¹, providing strong statistical evidence that the metabolite degrades at a rate significantly above zero.

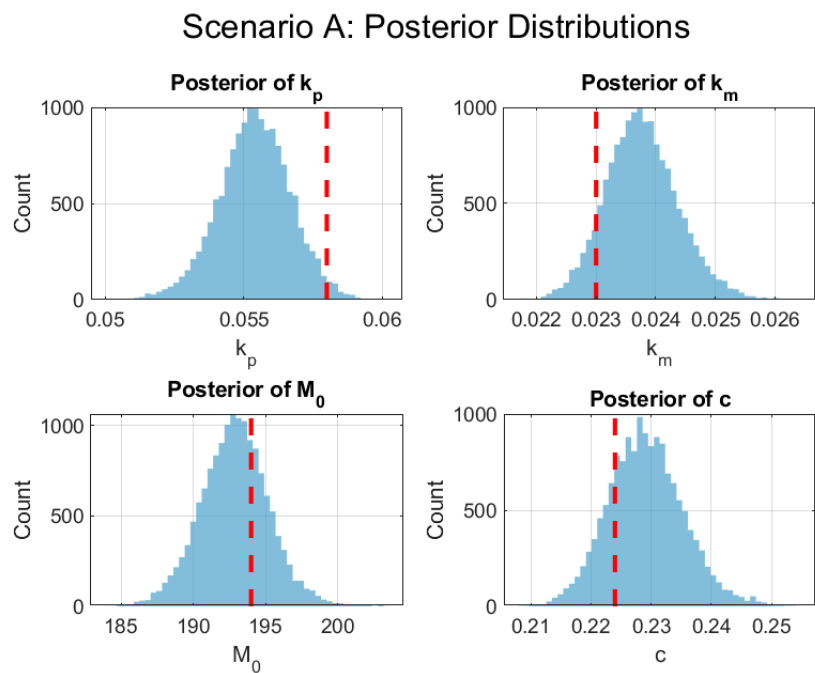


Figure 2. Marginal posterior histograms for k_p , k_m , M_0 , and c under Scenario A. Red dashed lines indicate true parameter values.

Predictive uncertainty bands (Figure 3) show that the 90% credible interval captures virtually all observations. The Parent compartment is well-constrained throughout the time horizon; the Metabolite interval is wider near peak concentration (~day 50) due to joint sensitivity to both k_p and k_m .

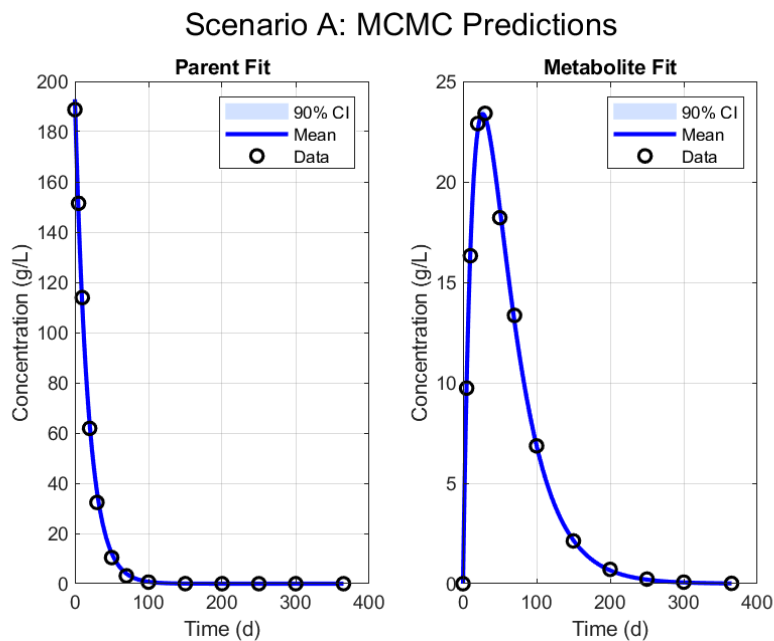


Figure 3. Posterior predictive mean (blue line) and 90% credible interval (shaded) overlaid on observed data for Parent (left) and Metabolite (right) under Scenario A.

4.2 Scenario B — Zero Metabolite Rate ($k_m = 0$)

When the true k_m is zero, the data contain no information to distinguish metabolite formation from metabolite persistence. The posterior of k_m (Figure 4) is strongly right-skewed and piles up near zero, as expected under the non-negativity constraint. The MCMC acceptance rate was 15.8%, reflecting slower mixing near the parameter boundary.

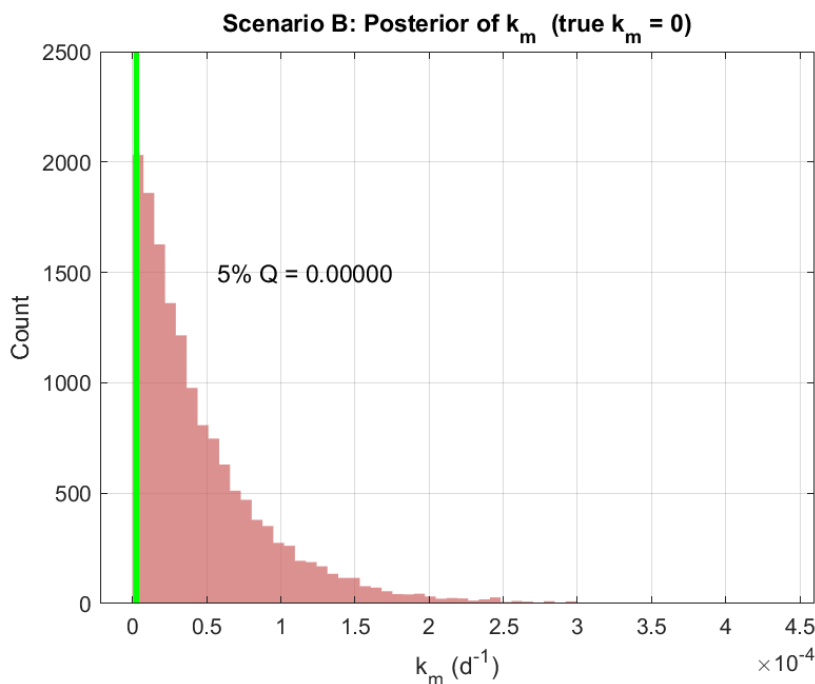


Figure 4. Posterior histogram of k_m under Scenario B. The green line marks the 5th-percentile quantile ($2.55 \times 10^{-6} \text{ d}^{-1}$), confirming negligible probability mass above any practically meaningful degradation rate.

The 5th-percentile of the k_m posterior is $2.55 \times 10^{-6} \text{ d}^{-1}$ — indistinguishable from zero at any practical level of precision. This provides the regulatory criterion: we cannot demonstrate degradation of the metabolite at any detectable rate.

4.3 Comparison: MCMC vs. Normal (Laplace) Approximation

Figure 5 compares the MCMC posterior of k_m (Scenario B) with the Gaussian approximation centred at the MAP estimate. The normal approximation — obtained from the numerical Hessian of the negative log-posterior via `fmincon` — predicts a 5th-percentile of -0.0005 d^{-1} , a physically impossible negative value.

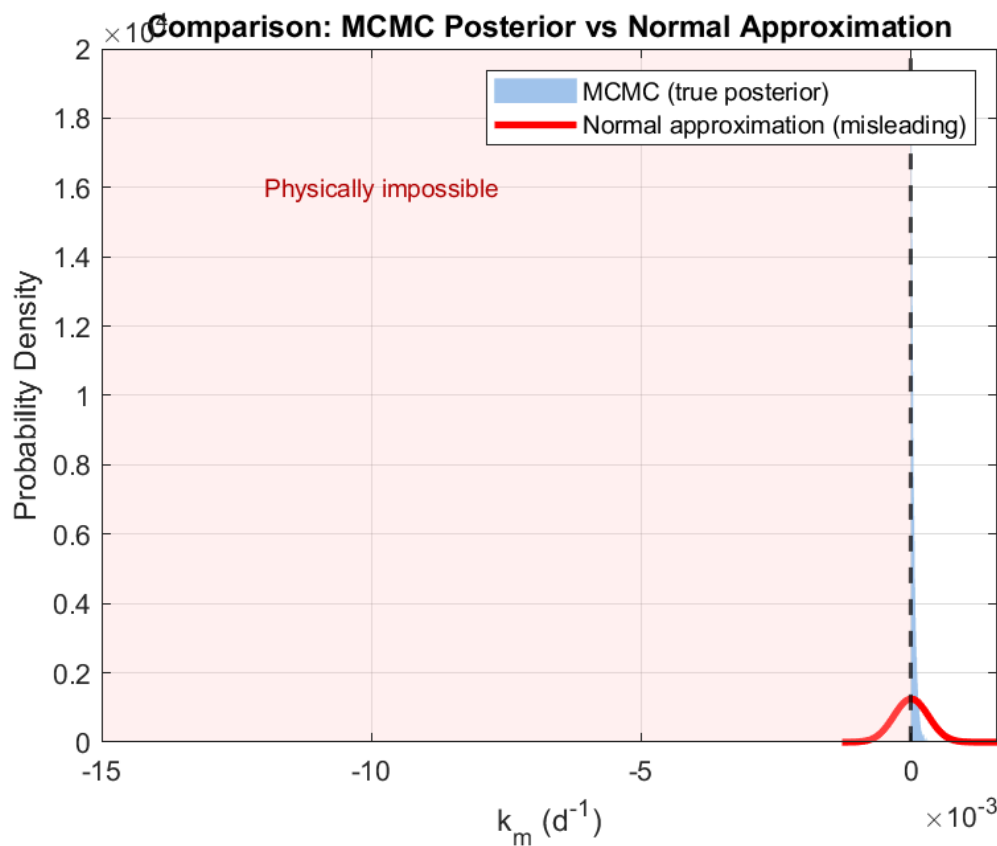


Figure 5. Normalised MCMC posterior (blue histogram) vs. the Gaussian approximation (red curve). The pink shaded region left of $k_m = 0$ represents the impossible tail predicted by the normal approximation.

This discrepancy arises because the MCMC correctly respects the hard physical constraint $k_m \geq 0$, while the symmetric Gaussian does not. The failure is especially severe when the MAP estimate lies close to a boundary — a generic pitfall in Laplace-based inference for constrained kinetic parameters.

4.4 Summary of Key Numerical Results

Table 3. Summary of MCMC diagnostic and inference results.

Metric	Scenario A	Scenario B
Acceptance Rate	42.9%	15.8%
5th Percentile of km	0.0228 d ⁻¹	2.55 × 10 ⁻⁶ d ⁻¹
Normal Approx. 5th Percentile	N/A	-0.0005 d ⁻¹ (impossible)
Interpretation	km confidently > 0	km indistinguishable from 0

6. Conclusions

This study demonstrates the application of Metropolis–Hastings MCMC for Bayesian parameter estimation of a two-compartment chemical degradation model. The key conclusions are:

- Scenario A (km > 0):** MCMC successfully recovered all true parameters with well-resolved, approximately Gaussian posteriors. The 5th-percentile of km (0.0228 d⁻¹) provides a statistically rigorous lower confidence bound supporting regulatory classification as a degrading metabolite.
- Scenario B (km = 0):** When the metabolite is persistent, MCMC correctly identifies km ≈ 0 with a strongly right-skewed posterior. The 5th-percentile (2.55 × 10⁻⁶ d⁻¹) is indistinguishable from zero, consistent with a regulatory conclusion of metabolite persistence.
- Superiority over Normal Approximation:** The Laplace approximation yields a negative 5th-percentile (-0.0005 d⁻¹) — physically impossible. MCMC respects the non-negativity constraint, demonstrating its fundamental advantage for boundary-constrained parameters.
- Methodological Value:** Full posterior sampling yields complete predictive distributions, enabling uncertainty-aware risk assessments that cannot be obtained from deterministic optimisation alone.

Appendix A — MATLAB Code Summary

The complete implementation is contained in a single MATLAB script (no additional toolboxes required beyond fmincon from the Optimization Toolbox). The key functional blocks are:

- model_func(theta, t):** Analytical solution for parent and metabolite concentrations. Includes the degenerate case (kp ≈ km) via a L’Hôpital limit.
- log_posterior(vec, t, y):** Evaluates the log-posterior density. Physical-bounds checking returns -Inf for infeasible proposals. Noise parameters are on the log scale.
- run_mcmc(init, n_iter, burn, thin, prop_sd, t, y):** Core MH sampler. Returns thinned post-burn-in samples and the overall acceptance rate.
- Normal approximation (Figure 5):** MAP found via fmincon; Hessian estimated by second-order finite differences; posterior covariance by matrix inversion.